Prometheus Lab:

ECSE 36 R1-D4 App and Swarm Robots

Advised By: Prof Joseph Vybihal

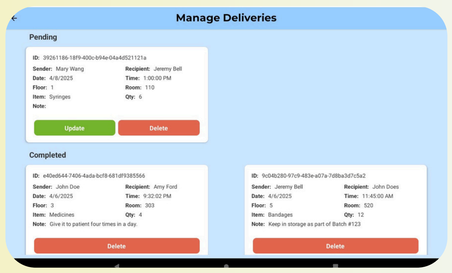
DP 36: Anjali Singhal, Joey Hershkop, Paolo Bizzari, Noorjahan Abonty

R1D4 Interface App

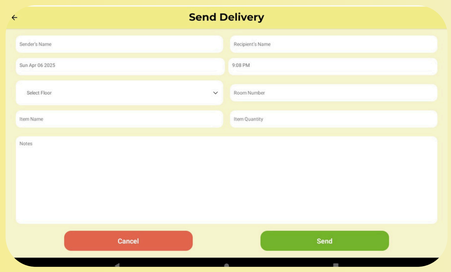
We created an intuitive touchscreen system for doctors and staff to use with the R1-D4 Hospital Robot. Users can place deliveries, receive pickups, and modify orders via our application, and also view the robot's real-time status via animated expressions. The app is built using React Native and JavaScript, running on an Android tablet. The backend utilizes an ESP32 running a Telnet server, and the ESP32's GPIO pins are used to control the R1D4 robot.



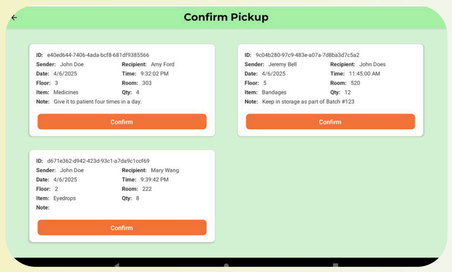
Home Screen



Manage Tasks



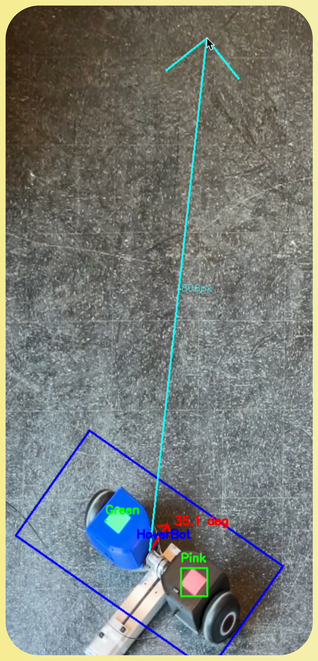
Send Delivery

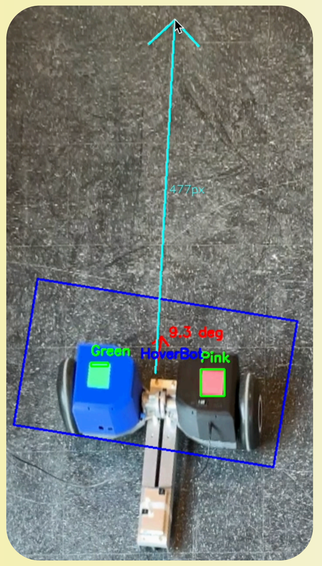


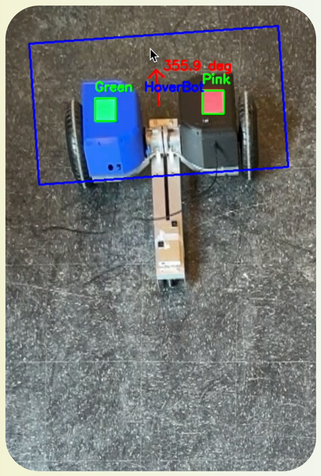
Confirm Pickup

HoverBot Navigation System

We created a computer vision navigation system for the HoverBot, one of the robots in the swarm robotics project. We use OpenCV and Continuity Camera to stream and process a parallax-corrected live video feed. Colour-based object detection tracks the position and orientation of the HoverBot. The user selects the robot and destination; the program then sends movement commands over a TCP server, using live feedback to update position information in real-time and maintain accuracy.





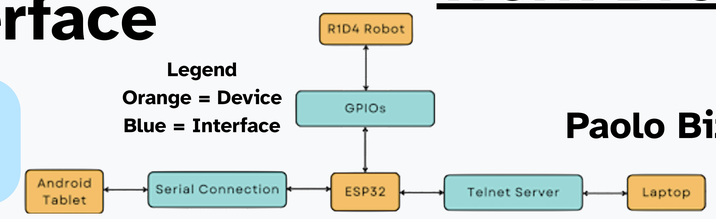


The destination spot is selected with the cursor. The robot turns counterclockwise to align itself and then moves forward, continuously adjusting its orientation until it reaches the destination.

Work Breakdown

R1D4 Interface

Designed, tested, and validated the communication system between the ESP32, the tablet, and the Telnet Server.



Paolo Bizzari

Addressed setup issues for the HoverBot. Spent significant time resolving hardware-related bugs. Solved these issues by replacing both motor controller boards and the 36V to 12V DC converter.

Contributed to frontend design. Worked on the ESP32 backend, getting the Telnet server operational. Helped establish the conceptual overview of database logic and user interface.

Joey Hershkop

Conceived and initiated the navigation system. Implemented object tracking, wireless camera feed, object labeling, parallax correction, aspect ratio correction, the logic/computation of the navigation algorithm, and the UI.

Set up the project repository and structure, designed the frontend, worked on the backend for all pages and app storage, added animations, and fixed bugs related to communication between the Telnet server and the app.



Anjali Singhal

Fixed bugs in the TCP server code. Worked on the logic/computation of the navigation algorithm, implemented sending API requests to the bot, and performed system testing on the bot.

Assisted in selecting and setting up the app framework by conducting a detailed comparison between building a website versus an Android app. Worked on the "Manage Tasks" and "Confirm Pickup" pages: completed their preliminary setup, accessed the global delivery array, and implemented the confirm button. Redesigned the task layout and added edit, delete, and confirm functionalities.

Noorjahan Abonty

Helped with the initial setup and wiring verification of the HoverBot hardware. Manually tested each pin by removing connections to observe expected motor behavior, and confirmed proper grounding and power connections. Followed hardware testing procedures to verify wheel responses and detect wiring faults.

Conclusion:

- **Gained experience with real-time embedded systems and Android app development.**
- **Learned the importance of early-stage time management in project execution.**
- **Developed teamwork, communication, planning, and management skills applicable to large engineering projects.**

Next Steps:

- **Integrate the R1D4 interface with the rest of the R1D4 robot hardware.**
- **Extend the computer vision system to support both the TankBot and the BoeBots.**
- **Integrate the new BoeBot into the swarm.**
- **Replace the battery on the HoverBot.**